NAU
NORTHERN ARIZONA UNIVERSITY

Project RAPTOR

Rocket Apogee Payload Transition to Optimal Return

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Date: September 18th, 2026

Project Overview

High Powered Rocket Module

- Solid propellant motor rocket
- Carries payload to the target altitude
  - Designing for a target of 1,500 – 2,500 ft. AGL (Above Ground Level)
- Supported by deployment module
- Electronics for telemetry, communication, and deployment

Deployable Drone Module

- Compact fixed wing aircraft that is housed within the rocket body
  - Wings must collapse for storage and open after apogee deployment
- Supported by controls module
- Electronics for telemetry, communication, controls and power

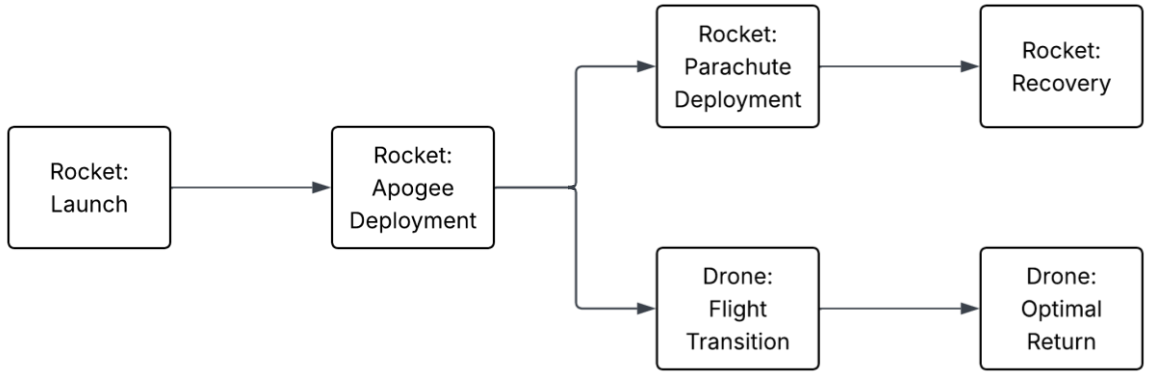


Figure 1: Concept of Operations

High Powered Rocket Module

- **Avionic Components**
  - Flight computer
  - Barometer
  - Inertial measurement unit
  - Communication
  - Telemetry
  - Deployment actuation

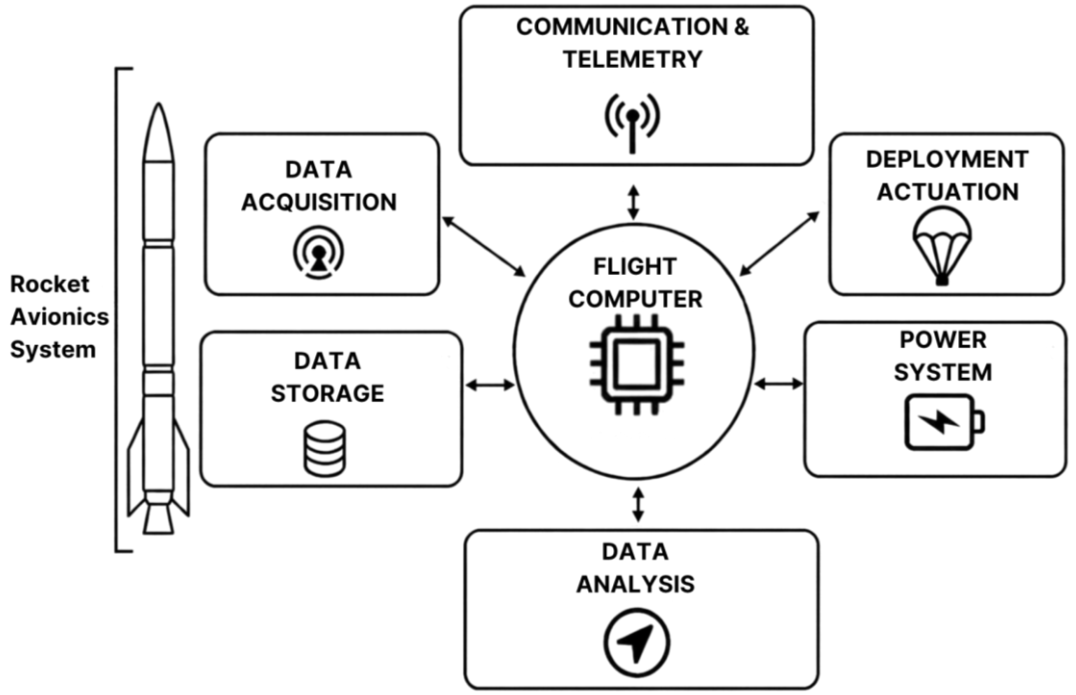


Figure 2: Rocket Avionic Components

- **Mission Phases**
  - 0) Pre-launch
  - 1) Launch
  - 1) Boost
  - 2) Coast
  - 2) Apogee
  - 3) Deployment
  - 3) Touchdown
  - 3) Recovery

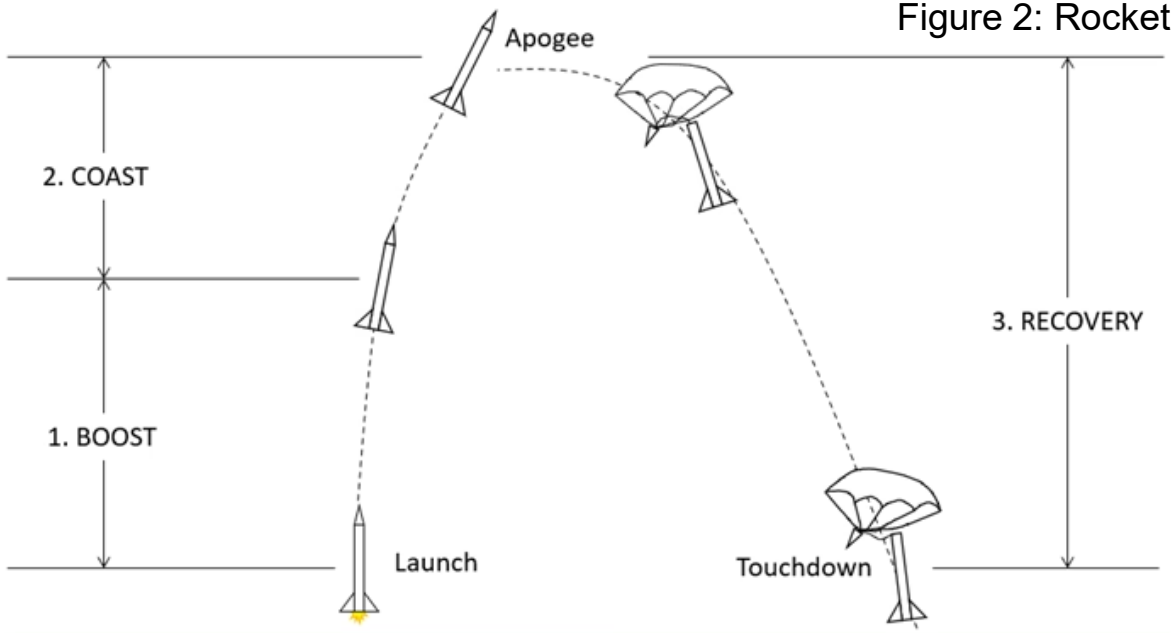


Figure 3: Rocket Flight Phases [1]

Deployable Drone Module

- **Avionic Components**
  - Flight computer
  - Electronic speed controller
  - Barometer
  - Inertial measurement unit
  - Communication
  - Telemetry
  - Control system

- **Mission Phases**
  - 1) Drone separation
  - 2) Stabilized flight
  - 3) Circular descent
  - 4) Parachute deployment
  - 5) Optimal landing

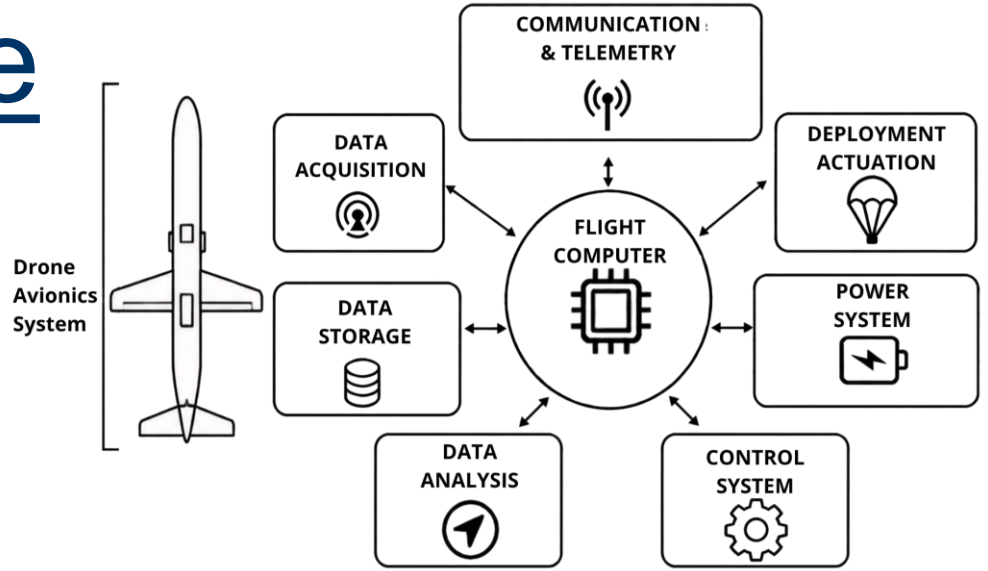


Figure 4: Drone Avionic Components

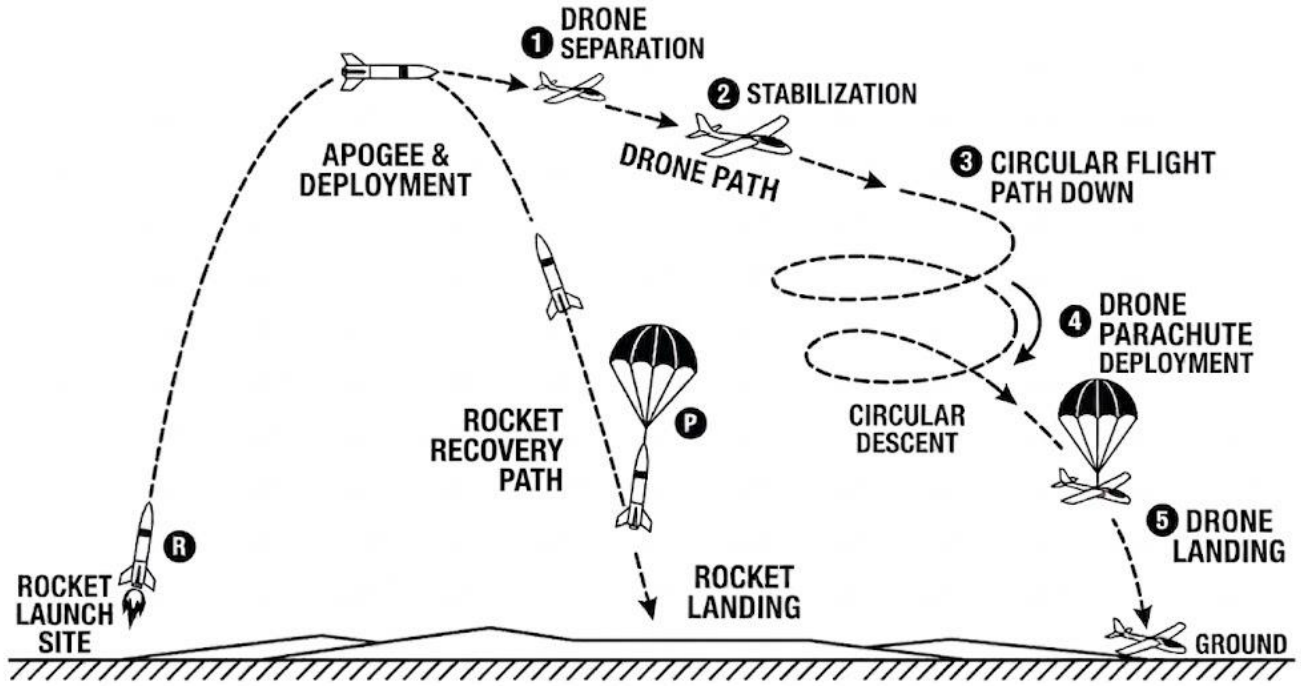


Figure 5: Drone Flight Phases

Requirements Overview

- **Integrated System:** Build a flight-ready rocket with an internally carried/integrated glider.
- **Target Apogee:** Reach 1,500–2,500 ft AGL; model trajectory, stability, and speed in RockSim.
- **Autonomous Deployment:** Detect apogee autonomously and safely eject the glider.
- **Stable Transition:** Demonstrate a clean, reliable shift from rocket release into glided flight.
- **Active Guidance:** Actively steer the glider back toward a designated recovery zone**.**
- **Telemetry Data:** Record flight performance, dynamics, and controls data for evaluation.
- **Mission Launch:** Execute an integrated flight and recovery at the TRA Phoenix launch site.
- **Safety Compliance:** Adhere strictly to Tripoli Rocketry Association (TRA) codes and ground testing.



Fig. 5 TRAPHX logo [2].

References

[1] “Solid Motor Rocket Propulsion | Science Project,” Science Buddies. Accessed: Sep. 15, 2026. [Online]. Available: https://www.sciencebuddies.org/science-fair-projects/project-ideas/Phys_p008/physics/model-rocket-propulsion

[2] Tripoli Phoenix, “TRAPHX Club Logo,” Tripoli Phoenix Club website. Accessed: Sep. 16, 2026. [Online]. Available: https://traphx.clubexpress.com/

Thank You